

using data from 1920 through 1935. Many other researchers have confirmed this approach in other markets and with additional data up to the present and all the way back to 1903 with absolute momentum and 1801 with relative momentum. With so many years of out-of-sample validation, we should not have to worry about data-snooping bias. The fact that a 12-month look-back works well with both forms of momentum serves as a cross-validation of 12 months as a good length for our look-back period.

With so much going for dual momentum, if you try to replace or modify this proven approach with something new, you face several potential problems. First is the multiple-comparisons hazard that comes from data mining when it becomes data snooping. If you look at enough different strategies, almost certainly a few of them will look attractive. However, this simply can be due to chance or luck. If you measure statistical significance at the 5% level, for example, and you test 20 or more strategies, you will likely end up with one that appears significant when it is not, since it can still occur 1 in 20 times by chance alone. One does not need to experiment extensively to get into trouble this way. In an aptly named study called “Pseudo-Mathematics and Financial Charlatanism: The Effects of Backtest Overfitting on Out-of-Sample Performance,” Bailey et al. (2014) state, “high performance is easily achievable after backtesting a relatively small number of alternative strategy configurations [I]nvestors can be easily misled into allocating capital to strategies that appear to be mathematically sound.

There are some infamous examples of what can happen when you rely only on data snooping to develop an explanatory model. Around 20 years ago, two researchers found they could explain 99% of the return on the S&P 500 Index using a multiple regression on butter production in Bangladesh, cheese production in the United States, and the number of sheep in the United States and Bangladesh. Those researchers still get inquiries asking where one can find data on Bangladesh butter production! There have been other studies linking stock market returns to the number of nine-year-old children in the United States, the length of women’s dress hemlines, and whether or not an American graces the cover of the annual *Sports Illustrated* swimsuit edition.¹

The same data-snooping problem exists after you choose a model and have to determine its parameters. With momentum, there is only one parameter: the look-back period. By now, researchers have validated it every